

# Data Quality & Cleaning Report

## Nutrition, Physical Activity and Obesity Dataset

Tableau Prep Cleaning Workflow - Week 2- Exercise – Individual

### 1. Dataset Description

| Field       | Details                                                                                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title       | Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS)                                                                                                                                                                                          |
| Source      | Centers for Disease Control and Prevention (CDC) – data.cdc.gov                                                                                                                                                                                                                         |
| URL         | <a href="https://data.cdc.gov/Nutrition-Physical-Activity-and-Obesity/Nutrition-Physical-Activity-and-Obesity-Behavioral/hn4x-zwk7/about_data">https://data.cdc.gov/Nutrition-Physical-Activity-and-Obesity/Nutrition-Physical-Activity-and-Obesity-Behavioral/hn4x-zwk7/about_data</a> |
| Time period | 2011 – 2023 (13 years)                                                                                                                                                                                                                                                                  |
| Records     | 106,260 rows × 29 columns (raw)                                                                                                                                                                                                                                                         |
| Format      | CSV                                                                                                                                                                                                                                                                                     |
| License     | Public domain – CDC open data                                                                                                                                                                                                                                                           |

#### What the dataset contains

The dataset tracks the prevalence of obesity, overweight, physical activity, and fruit/vegetable consumption among U.S. adults (18+) across all 50 states, the District of Columbia, U.S. territories (Puerto Rico, Guam, Virgin Islands), and a national aggregate row. Measurements are broken down by six demographic stratification dimensions: age group, sex, income bracket, educational attainment, race/ethnicity, and an unstratified total. Data come from the CDC's Behavioral Risk Factor Surveillance System (BRFSS), a telephone-based annual health survey.

#### Why this dataset was selected

This dataset satisfies all four FAIR principles in a verifiable way: it is openly accessible on a government portal, has persistent URLs and clear provenance, uses a tabular open format (CSV), and its variables are well-documented. It is also substantively relevant to public health research, large enough to illustrate a wide range of data quality issues (over 100,000 rows), and manageable enough to clean in a single Tableau Prep workflow. The combination of structural issues (column name typo, wrong data type) and semantic issues (masked missing values, redundant fields) makes it an ideal teaching dataset.

## 2. FAIR Compliance Summary

FAIR stands for Findable, Accessible, Interoperable, and Reusable. The table below summarises the dataset's compliance before and after cleaning.

| Principle            | Strengths                                           | Weaknesses addressed by cleaning                                             | Score |
|----------------------|-----------------------------------------------------|------------------------------------------------------------------------------|-------|
| <b>Findable</b>      | Named source (BRFSS), rich in-row metadata          | No global PID — outside scope of Prep                                        | 60%   |
| <b>Accessible</b>    | Open CSV, zero true nulls, public portal            | Unknown geo-coordinates converted to proper NaN                              | 80%   |
| <b>Interoperable</b> | Hierarchical IDs, consistent stratification pattern | Column name fixed; GeoLocation split into Lat/Lon; Education value completed | 75%   |
| <b>Reusable</b>      | 13-year coverage, CI + sample size included         | Unknown → NaN; redundant columns removed; Sample_Size cast to integer        | 70%   |

### 3. Identified Data Quality Issues

Six categories of issues were identified through programmatic profiling of the raw dataset using Python (pandas). The table below lists all issues, the quality dimension affected, and the number of records impacted.

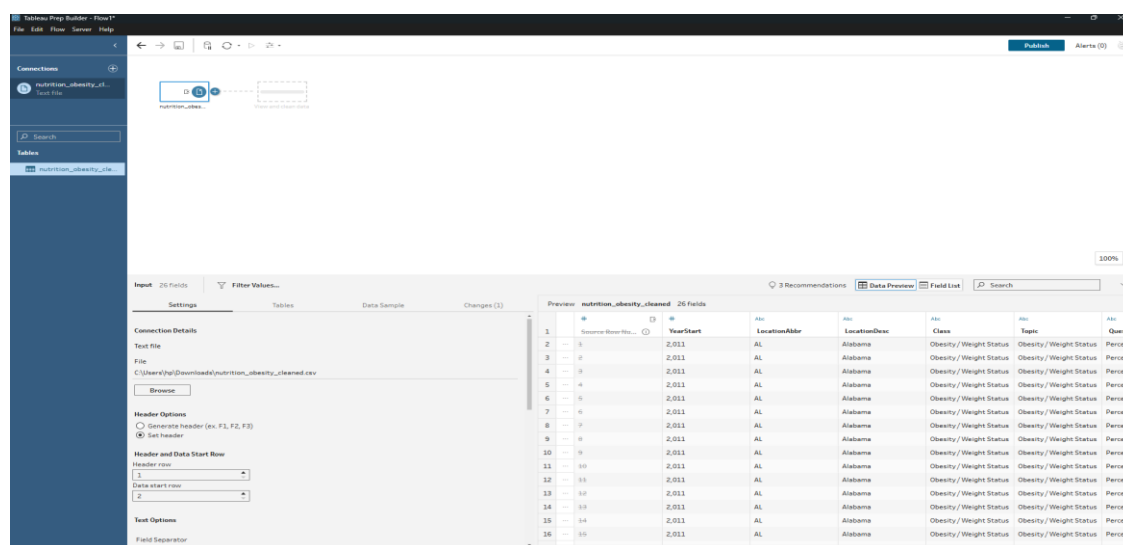
| # | Issue                                                      | Category                             | Rows affected                        | Description                                                                                                                                                                      |
|---|------------------------------------------------------------|--------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | "Unknown" masking missing values                           | Missing values                       | 75,900– 98,670 per column            | Five demographic columns use the string "Unknown" instead of null. Standard null-detection tools report 0 missing values, giving a false picture of completeness.                |
| 2 | GeoLocation as unparseable string + 1,932 "Unknown" values | Inconsistent format / Missing values | 106,260 format issue; 1,932 missing  | Coordinates are stored as the text string "(lat, lon)" rather than separate numeric columns. Additionally 1,932 rows contain the string "Unknown".                               |
| 3 | Trailing space in column name "High_Confidence_Limit"      | Inconsistent format                  | All 106,260 rows                     | The column has a trailing whitespace character in its name. Any code referencing it by name will raise a silent error or KeyError.                                               |
| 4 | Sample_Size stored as float64                              | Incorrect data type                  | All 106,260 rows                     | Sample sizes are whole-number counts of survey respondents. Storing them as floating-point numbers wastes memory and misrepresents their nature.                                 |
| 5 | Redundant/constant columns (5 columns)                     | Irrelevant fields                    | All 106,260 rows                     | Five columns carry zero analytical information: Data_Value_Alt (identical to Data_Value), Data_Value_Type, DataValueTypeID, Datasource, and YearEnd (always equal to YearStart). |
| 6 | Education value truncated at 30 characters                 | Invalid value                        | Rows where Education = that category | "Some college or technical sch" is truncated. The correct full value is "Some college or technical school", indicating a data export limitation.                                 |

## 4. Tableau Prep Cleaning Workflow

The workflow consists of five sequential steps applied in Tableau Prep Builder. Each step is described below, along with the exact operation used and the rationale.

### Step 1 — Connect to source

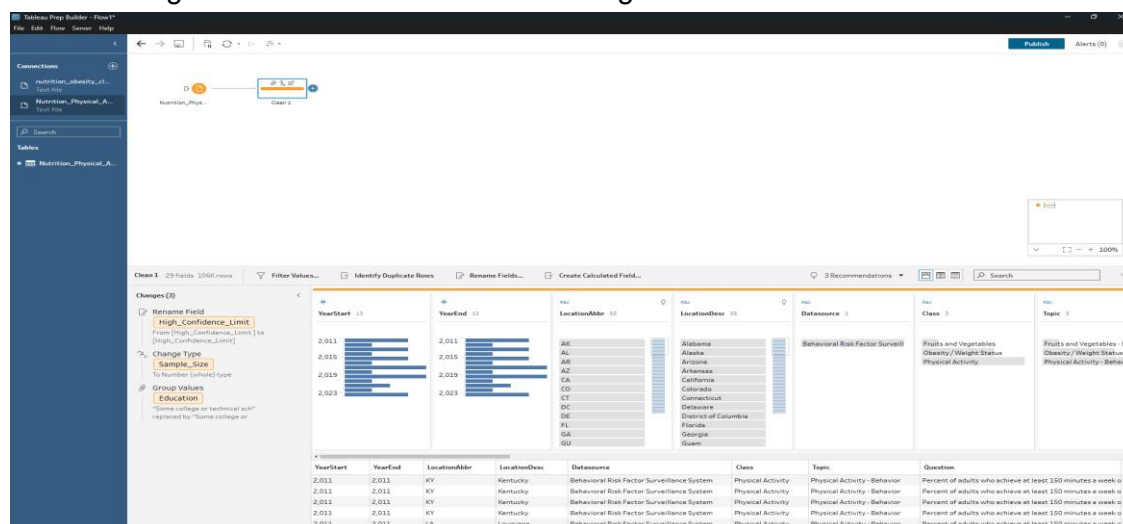
Add the raw CSV file as an input. In the Input pane, Tableau Prep will automatically profile each field and show data types, value distributions, and null counts. At this stage, note that Sample Size is detected as Number (Decimal) and that GeoLocation appears as a single string field.



### Step 2 — Clean step: fix column names and data types

Add a Clean Step node. Apply the following operations:

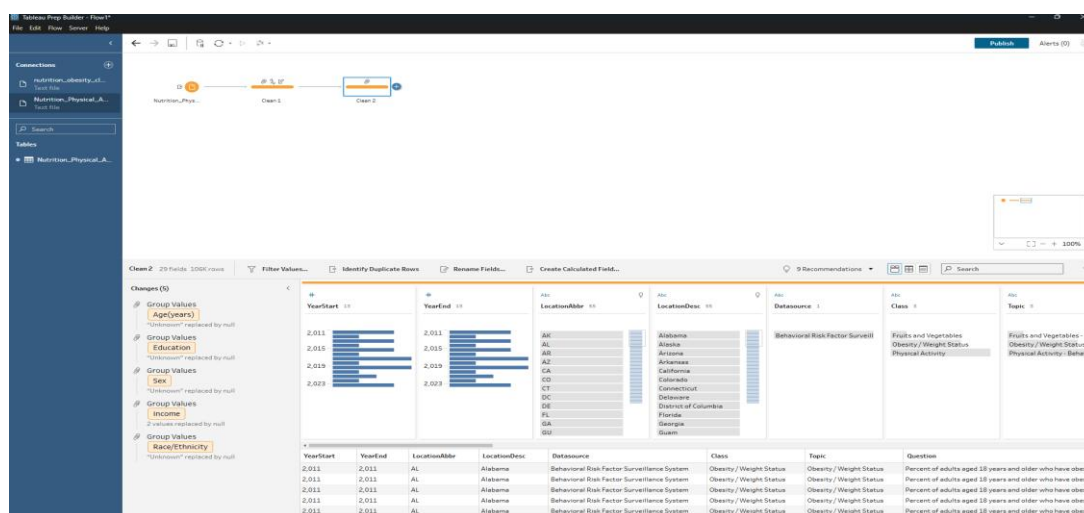
- Strip trailing whitespace from column names using Rename Field on "High\_Confidence\_Limit " → "High\_Confidence\_Limit"
- Change the data type of Sample\_Size from Number (Decimal) to Number (Whole) using the field type icon
- Complete the truncated Education value: use the Replace function to change "Some college or technical sch" to "Some college or technical school"



### Step 3 — Clean step: handle masked missing values

Add a second Clean Step. For each of the five demographic columns (Age(years), Education, Sex, Income, Race/Ethnicity), use the Filter or Replace function to replace the string "Unknown" and "Data not reported" with null. In Tableau Prep this is done by:

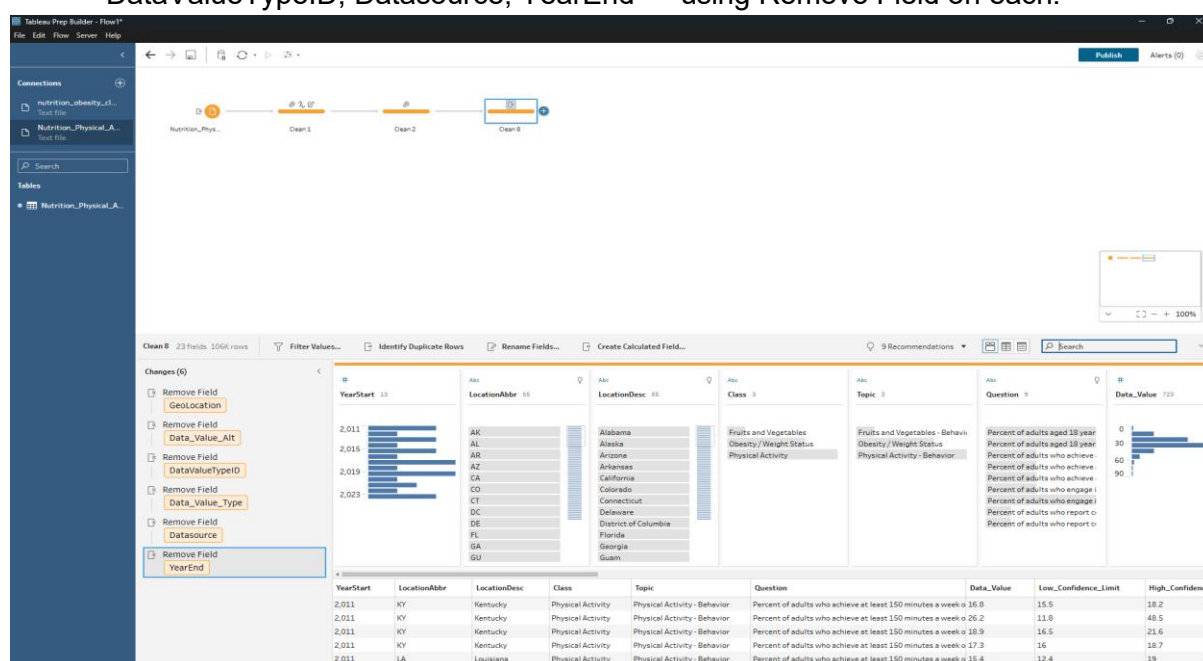
- Selecting the "Unknown" value in the field profile
- Right-clicking → Replace with Null
- Repeating for "Data not reported" in the Income field



### Step 4 — Clean step: split GeoLocation and remove redundant columns

Add a third Clean Step. Perform the following:

- Use Split Values on the GeoLocation field, splitting on the comma character. This creates GeoLocation-Split-1 (latitude) and GeoLocation-Split-2 (longitude). Rename them to Latitude and Longitude and change their types to Number (Decimal).
- Remove the original GeoLocation column using Remove Field.
- Remove the five redundant columns: Data\_Value\_Alt, Data\_Value\_Type, DataValueTypeID, Datasource, YearEnd — using Remove Field on each.



## Step 5 — Output step

Add an Output node at the end of the flow. Set the output format to CSV and name the file "nutrition\_obesity\_cleaned.csv". Run the flow to generate the cleaned file.

The screenshot shows the Tableau Prep interface with a flow consisting of five nodes: Nutrition\_Phys..., Clean 1, Clean 2, Clean 3, and Clean 4, followed by an Output node. The Output node is selected, and the 'Output' pane is open, showing a table of 24 fields. The 'Save output to' section is set to 'File', and the 'Output type' is 'Comma Separated Values (.csv)'. The 'Write Options' section is set to 'Full refresh' and 'Create table'. The 'Run Flow' button is visible at the bottom.

| Is Duplicate Row? | YearStart | LocationAbb | LocationDesc  | Class                   | Topic                        | Question                                                     | Data_Value | Low_Co |
|-------------------|-----------|-------------|---------------|-------------------------|------------------------------|--------------------------------------------------------------|------------|--------|
| Unique            | 2,011     | AL          | Alabama       | Obesity / Weight Status | Obesity / Weight Status      | Percent of adults aged 18 years and older who have obesity   | 35.2       | 30.7   |
| Unique            | 2,011     | AR          | Arkansas      | Obesity / Weight Status | Obesity / Weight Status      | Percent of adults aged 18 years and older who have obesity   | 31.2       | 28.5   |
| Unique            | 2,011     | FL          | Florida       | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in muscle-strengthening activ   | 27.2       | 23.8   |
| Unique            | 2,011     | IN          | Indiana       | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in muscle-strengthening activ   | 22.7       | 20.1   |
| Unique            | 2,011     | KY          | Kentucky      | Physical Activity       | Physical Activity - Behavior | Percent of adults who achieve at least 300 minutes a week o  | 30.8       | 26.5   |
| Unique            | 2,011     | OR          | Oregon        | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in muscle-strengthening activ   | 30.9       | 29.3   |
| Unique            | 2,011     | TN          | Tennessee     | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in muscle-strengthening activ   | 19.1       | 13.5   |
| Unique            | 2,011     | US          | National      | Physical Activity       | Physical Activity - Behavior | Percent of adults who achieve at least 150 minutes a week o  | 53.9       | 53.6   |
| Unique            | 2,012     | IA          | Iowa          | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in no leisure-time physical act | 29.6       | 25.8   |
| Unique            | 2,012     | NH          | New Hampshire | Obesity / Weight Status | Obesity / Weight Status      | Percent of adults aged 18 years and older who have an overs  | 34.8       | 33     |
| Unique            | 2,012     | NY          | New York      | Obesity / Weight Status | Obesity / Weight Status      | Percent of adults aged 18 years and older who have an overs  | 35.8       | 31.5   |
| Unique            | 2,013     | AR          | Arkansas      | Physical Activity       | Physical Activity - Behavior | Percent of adults who achieve at least 150 minutes a week o  | 44.9       | 41     |
| Unique            | 2,013     | AZ          | Arizona       | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in no leisure-time physical act | 14.8       | 12.1   |
| Unique            | 2,013     | CO          | Colorado      | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in no leisure-time physical act | 22.8       | 17.5   |
| Unique            | 2,013     | DE          | Delaware      | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in no leisure-time physical act | 34.5       | 30.1   |
| Unique            | 2,013     | FL          | Florida       | Obesity / Weight Status | Obesity / Weight Status      | Percent of adults aged 18 years and older who have obesity   | 21.8       | 20.3   |
| Unique            | 2,013     | KS          | Kansas        | Physical Activity       | Physical Activity - Behavior | Percent of adults who engage in muscle-strengthening activ   | 32.3       | 27.9   |
| Unique            | 2,013     | LA          | Louisiana     | Physical Activity       | Physical Activity - Behavior | Percent of adults who achieve at least 150 minutes a week o  | 21.2       | 15.5   |

## 5. Data Quality Dimensions

Each cleaning step maps to one or more of the standard data quality dimensions. The table below shows this mapping explicitly.

| Cleaning action                                   | Quality dimension(s)          | How it improves quality                                                                                                                                                               |
|---------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Replace “Unknown” → null in 5 demographic columns | Completeness, Accuracy        | True missing values are now represented correctly. Statistical analyses (mean, groupby, isnull) will behave as expected. Completeness metrics now reflect the real state of the data. |
| Fix trailing whitespace in column name            | Consistency, Usability        | Column can now be referenced by its exact logical name without silent failures. The dataset schema is internally consistent.                                                          |
| Sample_Size: float64 → integer                    | Accuracy, Validity            | The data type now correctly reflects the semantic nature of the field (a count of people). Integer arithmetic and aggregations will behave correctly.                                 |
| Split GeoLocation into Latitude & Longitude       | Consistency, Interoperability | Coordinates are now machine-readable as separate numeric fields, compatible with GIS tools, mapping libraries, and spatial joins without preprocessing.                               |
| Remove 5 redundant columns                        | Relevance, Usability          | Removing constant and duplicate columns reduces noise, file size, and cognitive overhead for analysts. The remaining 25 columns all carry unique, analytical information.             |
| Complete truncated Education value                | Accuracy, Validity            | The full category name is now present, preventing misidentification and ensuring correct grouping in analyses and visualisations.                                                     |

## 6. Summary

### Main problems found

- Masked missing values in 5 demographic columns using the string “Unknown” (up to 98,670 rows per column)
- GeoLocation stored as an unparseable string; 1,932 rows contained “Unknown” instead of coordinates
- Trailing whitespace in the column name “High\_Confidence\_Limit ” — affecting all 106,260 rows
- Sample\_Size incorrectly typed as float64 instead of integer
- Five redundant or constant columns carrying no analytical value
- One Education category value truncated to 30 characters

## Steps applied

1. Input step: loaded raw CSV into Tableau Prep
2. Clean step 1: renamed column (strip whitespace), changed Sample\_Size type to integer, fixed truncated Education value
3. Clean step 2: replaced “Unknown” and “Data not reported” with null across 5 demographic columns
4. Clean step 3: split GeoLocation into Latitude/Longitude, removed original GeoLocation column, removed 5 redundant columns
5. Output step: exported cleaned dataset as nutrition\_obesity\_cleaned.csv

## Improvements achieved

| Metric                                       | Before                         | After                                     |
|----------------------------------------------|--------------------------------|-------------------------------------------|
| Number of columns                            | 29                             | 25 (4 removed, GeoLocation replaced by 2) |
| Detectable null values in demographic fields | 0 (all hidden as “Unknown”)    | Correctly represented as null/NaN         |
| GeoLocation usability                        | String, not parseable directly | Two numeric columns (Latitude, Longitude) |
| Column name errors                           | 1 (trailing space)             | 0                                         |
| Sample_Size data type                        | float64                        | Int64 (integer)                           |
| Truncated category values                    | 1 (Education)                  | 0                                         |

## Remaining limitations

- No persistent identifier (DOI/PID): The dataset still lacks a globally unique, resolvable identifier embedded in the file. This is a FAIR findability gap that cannot be resolved within Tableau Prep.
- No license field: There is no machine-readable license statement in the data. Users must visit the CDC portal to confirm terms of use.
- Semantic masking of missingness is by design: “Unknown” was used because each row is stratified on only one demographic dimension at a time — the other four are genuinely “not applicable” rather than “not collected”. Converting all to null loses this nuance. A production pipeline should distinguish between “not applicable” and “missing”.
- Outliers not removed: Statistical outliers exist in several question groups (e.g. obesity prevalence values at the tails of sub-population distributions). These were not removed as they may represent valid extreme values for specific demographic sub-groups.
- No data dictionary: The cleaned file still lacks an accompanying codebook documenting the meaning and units of each field. Units for Data\_Value (implied percentage) are not stated.



## References

Centers for Disease Control and Prevention. (2024). Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System [Dataset]. data.cdc.gov. Retrieved April 2026, from [https://data.cdc.gov/Nutrition-Physical-Activity-and-Obesity/Nutrition-Physical-Activity-and-Obesity-Behavioral/hn4x-zwk7/about\\_data](https://data.cdc.gov/Nutrition-Physical-Activity-and-Obesity/Nutrition-Physical-Activity-and-Obesity-Behavioral/hn4x-zwk7/about_data)

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3, 160018. <https://doi.org/10.1038/sdata.2016.18>